Lab 1 – Requirements Engineering & UML Use-Case Modelling

Problem Statement #02: Automated Rubric Assignment Evaluator

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall automatically queue uploaded student project ZIP files, execute pre-configured unit test suites, and generate an itemized rubric score breakdown.	High	Pass: The rubric breakdown is rendered with test results within 10 seconds of submission. Fail: the file queue stalls or the rubric tally is invalid.	Core workflow directly automates batch evaluation and scoring.	Given
FR-002	Functional	The system shall validate each uploaded project ZIP file for required structure, supported files, and archive integrity before placing it in the evaluation queue.	High	Pass: A valid submission is accepted and queued; an invalid/corrupted submission is rejected with a specific error and is not executed.	Prevents invalid submissions from reaching the execution environment.	
FR-003	Functional	The system shall automatically assign each submitted project to the configured number of eligible peer reviewers without assigning a student their own project.	High	Pass: In a test batch, every project receives the configured number of eligible reviewers and no reviewer is assigned their own submission. Fail: any project is unassigned or self-assigned.	Removes manual distribution overhead and avoids biased/self-review assignments.	
FR-004	Functional	The system shall allow the faculty evaluator to configure rubric criteria, criterion weights, and the unit-test suite for an assignment.	Medium	Pass: The evaluator can save the configuration, and the next evaluation uses the saved criteria, weights, and test suite. Fail: any saved setting is ignored during evaluation.	Faculty must control how submissions are tested and scored.	
FR-005	Functional	The system shall allow students, peer reviewers, and faculty evaluators to view the appropriate evaluation results and peer-review status for submitted projects.	Medium	Pass: After evaluation, an authorized user can view the applicable score/results and review status; unauthorized users cannot view another student's results.	Provides transparent access to outcomes while respecting access boundaries.	
NFR-001	Nonfunctional	The system shall handle up to 100 concurrent project submissions without dropping queue items or exceeding a 1.5 GB memory footprint.	High	Pass: Benchmarking under 100 concurrent submissions shows no dropped queue items and memory usage remains at or below 1.5 GB.	Ensures reliable performance during peak submission periods.	Given
NFR-002	Nonfunctional	The system shall execute submitted code in an isolated sandbox that prevents access to protected host files and unauthorized network resources.	High	Pass: Security tests show that submitted programs cannot read protected host files or access unauthorized network resources. Fail: any protected resource is accessible.	Protects the evaluation infrastructure from untrusted student code.	

Actors and Use Cases

Actors: Student, Faculty Evaluator, Peer Reviewer, Execution Sandbox.

- UC-01 Upload Project
- UC-02 Evaluate Student Project
- UC-03 Configure Assignment
- UC-04 Assign Peer Reviews
- UC-05 Submit Peer Review
- UC-06 View Evaluation Results
- UC-07 Validate Submission
- UC-08 Run Unit Tests
- UC-09 Generate Rubric Score

Relationship notes: UC-02 «include» UC-07, UC-08, and UC-09. UC-04 «extend» UC-02 when peer-review assignment is enabled after evaluation.